

Ahmed Metawea

AI & Data Science Engineer

✉ ahmed.metawea001@gmail.com ☎ (+20)1553441966 📍 Cairo, Egypt 🔗 LinkedIn 🐙 Github

SUMMARY

AI Engineer with a strong foundation in Machine Learning, Computer Vision, and NLP. I am driven by a genuine passion for continuous learning and staying at the forefront of emerging technologies — translating that curiosity into practical, end-to-end solutions across diverse domains. I bring both academic depth and hands-on project experience to every challenge I take on.

EDUCATION

Bachelor of Artificial Intelligence

07/2025

Faculty of Artificial Intelligence, Kafr Elsheikh University

Overall Grade: Very Good

SKILLS

Programming Languages & Frameworks

Python, Flask, FastAPI, SQL, Streamlit

Machine Learning & AI Domains

Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision

Tools & Technologies

Scikit Learn, PyTorch, TensorFlow, OpenCV, FAISS, Amazon SageMaker, AWS Machine Learning Services, LangChain, LangGraph, Docker

Soft Skills

Teamwork, Effective Communication, Problem Solving

PROJECTS

Safe Skin – AI-Driven Skin Cancer Detection Platform (Graduation Project) 🔗

- **Model Development** : Fine-tuned a Vision Transformer (ViT) model on the HAM10000 dataset, achieving **96.95% validation accuracy** across 7 lesion classes.
- **Data Engineering** : Built a robust preprocessing pipeline using data augmentation and weighted cross-entropy loss to solve severe class imbalance.
- **Deployment & Integration** : Deployed the model into a full-stack Flask web app, integrating Google Gemini for an AI chatbot and Google Maps API for locating clinics.

Sentinel CV — Multi-Capability Vision Monitoring Platform | Streamlit, YOLOv11, Keras, FaceNet 🔗

- Built a modular computer-vision platform unifying **5 capabilities (face recognition, phone/violence/fall/cheating detection)** under one config-driven architecture
- Implemented lazy model loading and LRU caching; wrote **54 tests** and containerized the app with Docker.

Arabic Dialect-to-MSA Neural Machine Translation (NLP) 🔗

- Fine-tuned transformer models (AraT5, mT5, NLLB, AraBART) on the MADAR parallel corpus (~51K sentence pairs across Egyptian, Gulf, and Levantine dialects)
- Achieving **71.56 BLEU** with **AraT5**; built and deployed a Flask web app serving the best model via REST API.

DAAS (Data Analysis Automation System) — Multi-Agent BI Platform 🔗

- Built a full-stack, multi-agent BI platform (**FastAPI + Next.js, PostgreSQL, LangGraph**) letting SME/e-commerce owners ingest data from files, Google Sheets, or a linked database and get automated cleaning, analytics, forecasting, marketing, and churn prediction.
- Designed **10 specialized agents** covering data cleaning (HITL), RFM segmentation, back-tested forecasting, churn prediction (**AUC≈0.93**), root-cause analysis, and scheduled autonomous monitoring with alerts.
- Built a deterministic-first architecture — all KPIs/forecasts/scores computed via pandas/scikit-learn/Prophet, with a multi-provider LLM layer (Groq→Anthropic→OpenAI→OpenRouter) used only to narrate — eliminating hallucinated figures.

SmartLab – AI Medical Laboratory Analysis System (Team Project)

- Built an end-to-end pipeline that scans lab reports via OCR, classifies report type using NLP, and routes to purpose-built ML models for CBC, Liver, Kidney, and Heart disease analysis.
- Developed ensemble models (Stacking Classifier, Random Forest) achieving up to **98.76% accuracy** and **ROC-AUC: 0.996** across four diagnostic domains.
- Integrated an LLM-powered chatbot (nvidia/nemotron-nano-12b-v2-vl) to explain results.
- **Personal contributions:** OCR pipeline implementation and Liver Function ML model.

Multilingual Text Preprocessing (End to End NLP & MLOps)

- Engineered a modular, dual-language text preprocessing engine supporting English and Arabic processing pipelines using FastAPI.*
- **Arabic NLP Engineering:** Implemented advanced Arabic text normalization using PyArabic, Tashaphyne, and ISRIStemmer, handling tasks like Tashkeel/Tatweel removal and root extraction.
 - **English NLP Engineering:** Integrated spaCy and NLTK for rule-based tokenization, intelligent stopword preservation (negations), and lemmatization.
 - **Production Deployment:** Containerized the entire application and its static web GUI into a single Docker image to ensure consistent environment provisioning and deployment readiness.
 - **API Design:** Developed configurable REST API endpoints using Pydantic schemas, allowing dynamic pipeline modification via JSON payloads.

TRAININGS & COURSES

Digilians Initiative Specialized Diploma (9-month program) <i>AI and Data Science Track</i>	12/2025 – 08/2026 Military Academy
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- IBM AI Engineering Professional Certificate (Coursera)** 
- Mastered training, fine-tuning, and deploying deep learning models and LLMs using Keras, PyTorch, and TensorFlow.
 - Built NLP applications leveraging LangChain, Hugging Face, BERT, and GPT/LLaMA architectures.
 - Implemented a capstone project using RAG and LangChain.

- AWS Certified Machine Learning Engineer – Associate (MLA-C01)** 
- Validates expertise in developing, deploying, optimizing, and maintaining machine learning solutions using AWS services.

- Huawei HCIA-AI (Huawei Certified ICT Associate – Artificial Intelligence)** 
- Covered core AI/ML fundamentals, deep learning frameworks, and Huawei's AI development ecosystem .

- Machine Learning Specialization (by Andrew Ng)**
- Mastered the fundamentals of machine learning, including Supervised and Unsupervised learning, Gradient Descent, and Regularization.

AI & Machine Learning with Quantum Computing Foundations 	02/2024
<i>Ministry of Youth and Sports (Shabab Mobtakeron Initiative)</i>	

- Completed a comprehensive program covering core Artificial Intelligence and Machine Learning concepts.
- Explored the theoretical foundations of Quantum Computing and its potential impact on advanced computing technologies.

- Artificial Intelligence A-Z 2025: Agentic AI, Gen AI, and RL(Udemy)** 
- Gained comprehensive knowledge in Reinforcement Learning and Generative AI.
 - Implemented practical AI agents and explored the architecture of Large Language Models (LLMs).

ACTIVITIES

Egyptian Collegiate Programming Contest (ECPC) Contestant	2023 Alexandria
• Solved complex algorithmic and mathematical problems using advanced Data Structures and Algorithms under strict time constraints.	

LANGUAGES

Arabic Native	English Advanced (Reading and Writing)
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